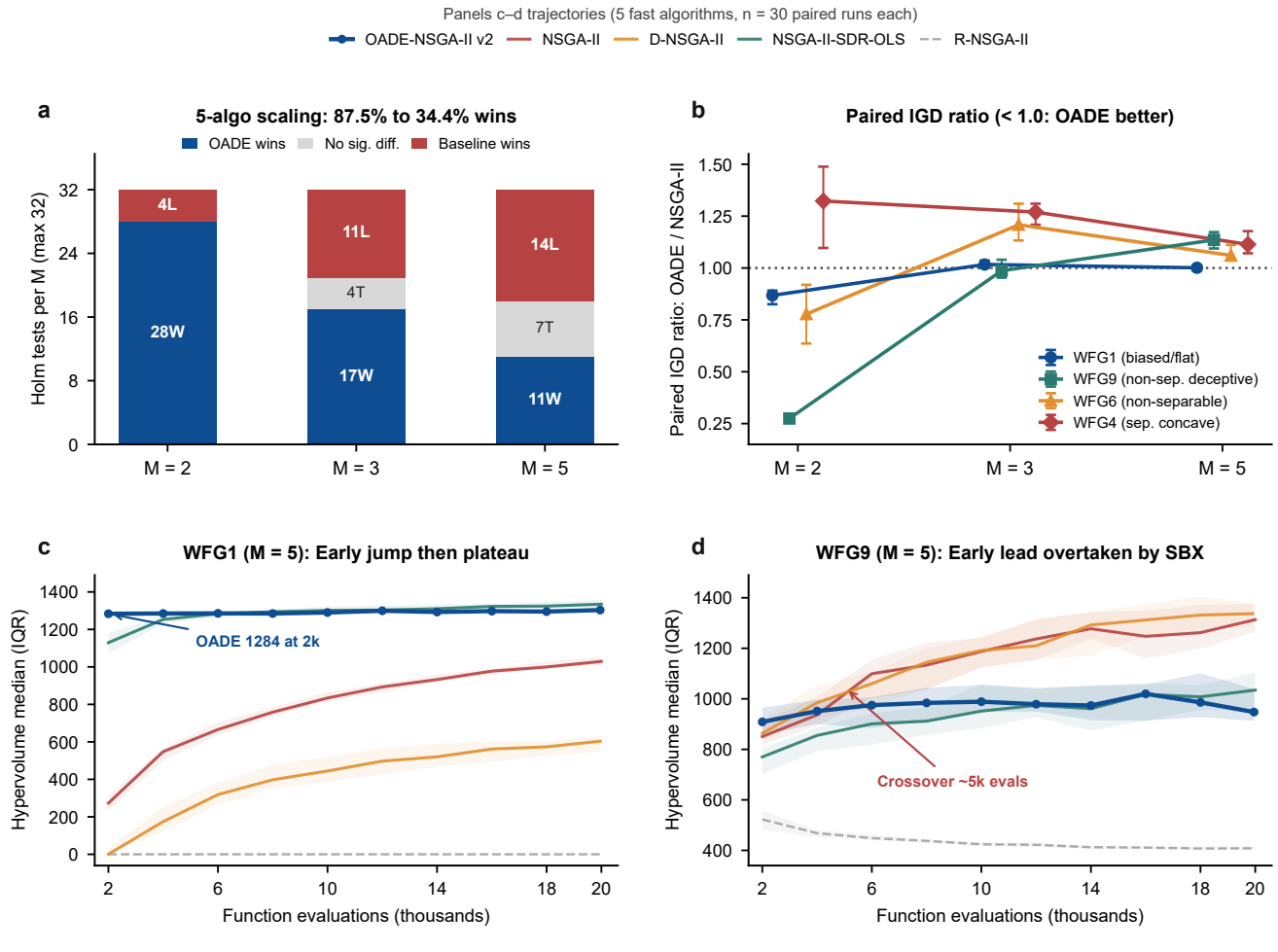

Panels c–d trajectories (5 fast algorithms, n = 30 paired runs each)
OADE-NSGA-II v2
NSGA-II
D-NSGA-II
NSGA-II-SDR-OLS
R-NSGA-II
a
5-algo scaling: 87.5% to 34.4% wins
OADE wins
No sig. diff.
Baseline wins
Holm tests per M (max 32)
4L
28W
11L
4T
17W
14L
7T
11W
M = 2
M = 3
M = 5
b
Paired IGD ratio (< 1.0: OADE better)
Paired IGD ratio: OADE / NSGA-II
WFG1 (biased/flat)
WFG9 (non-sep. deceptive)
WFG6 (non-separable)
WFG4 (sep. concave)
c
WFG1 (M = 5): Early jump then plateau
Hypervolume median (IQR)
OADE 1284 at 2k
Function evaluations (thousands)
d
WFG9 (M = 5): Early lead overtaken by SBX
Crossover ~5k evals